CPP NOTES -DAY 37

Sequence Containers

Store elements in a linear sequence. The order of elements is maintained as per the sequence in which they were inserted. Eg: vector(dynamic array) , deque(double-ended queue), list(doubly linked-list), forward_list(singly linked list), array, string.

Associative Containers

Store elements in a way that allows fast retrieval based on keys. The elements are ordered by keys.eg:

1. Set(collection of unique elements, ordered by keys),
2. Multiset(collection of elem,allow dupl, ordered by keys),
3. Map(collection of key-value pair with unique keys, ordered by keys),
4. Multimap(collection of key-value pairs, allow dup keys, ordered by keys).

- Provide methods for fast search, insertion and deletion.
- Suitable for scenarios like lookup search

Unordered Containers

Similar to associative containers but do not maintain any specific order. They provide average constant time complexity for search, insert and delete operations.eg:

1. Unordered_set(collection of unique elements, unordered)
2. Unordered_multiset(collection of elem,allow dupl, unorderd)
3. Unordered_map(collection of key-value pair with unique keys, unordered)
4. Unordered_multimap(collection of key-value pairs, allow dup keys, unordered)

Use hash tables internally for fast operations

FYI:

Shring_to_fit() – minimizes memory usage(string)

- *Can I allocate a zero sized for an array?- in cpp it does not allow, throws compilation error, but with vector we can create.*
  Eg: std::vector<int> v(0); // OK, empty vector